

A stochastic volatility model parameter calibration routine

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Abstract

We specify the routine that calibrates the five Heston (1993) parameters $\Phi = (\theta, \kappa, \eta, \rho, v_0)$ to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot S_{ref} , the moneyness-normalised and grid-snapped strike K^* , the surface-coverage requirements, the in-engine relative-price objective, the implied-volatility (“IV-space”) selection and acceptance gate, and the boundary-pegging rejection rule.

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1 The model

Under the risk-neutral measure, with constant risk-free rate r and continuous dividend yield g , the log-price $X_t = \ln S_t$ and instantaneous variance v_t of the SPX follow the Heston (1993) dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where W, B are independent Brownian motions, so $\text{corr}(dX_t, dv_t) = \rho dt$. The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with $\theta > 0$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\eta > 0$ the volatility of variance (QuantLib's `sigma`), $\rho \in [-1, 1]$ the leverage correlation, and $v_0 > 0$ the initial variance. The Feller quantity

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit; $\mathcal{F}(\Phi) \geq 0$ guarantees $v_t > 0$ almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

1.1 Pricing operators

Fix a valuation with maturity $\tau > 0$ (in years), strike K , and type $w \in \{+1, -1\}$ (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (5)$$

Black price. With total standard deviation $\Sigma = \sigma\sqrt{\tau}$,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (6)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where $\mathcal{N}$ is the standard normal CDF and $F = F(S, \tau)$.

Heston price. The Heston model price $C_H(\Phi; S, K, \tau, w)$ is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (7)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for $j \in \{1, 2\}$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (8)$$

and characteristic functions $f_j = \exp(C_j + D_j v_0 + iu \ln F)$ given by the standard coefficients

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (9)$$

$$C_j = \frac{\kappa\theta}{\eta^2} \left[(b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (10)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

where $\mu_1 = \frac{1}{2}$, $\mu_2 = -\frac{1}{2}$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$. In code this integral is evaluated by QuantLib's **AnalyticHestonEngine** (which uses a branch-stable formulation of (8)); equations (7)–(8) define the operator it realises.

Implied volatility. The Black-implied volatility of a price C is the inverse of (6) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (11)$$

computed numerically (QuantLib **blackFormulaImpliedStdDev** on the forward (5), returning Σ , then $\sigma = \Sigma/\sqrt{\tau}$). This operator defines the IV-space residuals of §5.

2 Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day d , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by i , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (12)$$

being respectively the quote timestamp, strike, type $w_i \in \{+1, -1\}$, traded size, trade price, trade-implied volatility, the underlying level *at trade time* t_i , and the days to maturity. Maturities enter the engine as τ_i in years under Actual/365, i.e. $T_i = \tau_i/365$.

The day is rate-stamped with $r = r(d)$ and $g = g(d)$, the most recent risk-free and dividend rates on or before d (a flat-forward curve for each; under flat curves the evaluation date is immaterial to T_i , so it is set to the QuantLib “today”).

Pre-filter. We keep only economically usable, Heston-appropriate quotes:

$$\mathcal{D} = \{ i : \sigma_i > 0 \text{ and } \tau_i \geq \tau_{\min} \}. \quad (13)$$

The floor $\tau_{\min}$ (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive η, κ to Feller-violating extremes that pollute the pooled fit.

3 Handling intraday spot movement

A Heston calibration is anchored to a *single* spot S , yet the trades in $\mathcal{D}$ occur across an intraday range of underlying levels $\{S_i\}$. We reconcile the two with one reference spot and a moneyness re-striking.

Reference spot. Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (14)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (15)$$

A flagged day ($R > R_{\max}$) is still calibrated and written, but the normalisation below is recorded as strained (its S_{ref} should be treated with suspicion).

Moneyness normalisation and grid snap. Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width Δ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (16)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in K/S rather than in the absolute strike. Re-striking by (16) therefore re-centres the whole day onto S_{ref} so that trades executed at different S_i but the *same* moneyness share a common surface column. The grid snap collapses the continuum of K_i^* values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§8) nonetheless use each contract's *original* (S_i, K_i) , since the calibrated Φ is spot-independent.

4 Surface construction

From the normalised trades we build a partial $K^* \times \tau$ implied-volatility surface, selecting the most liquid, nearest-the-money quotes.

Maturity selection. For each distinct maturity τ present in $\mathcal{D}$, let $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$ be its traded volume. Keep the N_t highest-volume maturities,

$$\mathcal{T} = \text{top-}N_t \text{ maturities by } V(\tau). \quad (17)$$

Strike selection (per maturity, per wing). Within each kept maturity $\tau \in \mathcal{T}$, separate calls ($w_i = +1$, OTM with $K_i^* \gtrsim S_{\text{ref}}$) from puts ($w_i = -1$, OTM with $K_i^* \lesssim S_{\text{ref}}$) and keep the n_k nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{n_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{n_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (18)$$

Pivot to a surface. For each retained cell (k, τ) with $k \in \mathcal{K}_\tau$, set the surface entry to the implied volatility of the *latest* trade in that cell,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \min_{i: K_i^* = k, \tau_i = \tau} (-t_i) \quad (\text{i.e. } \max t_i), \quad (19)$$

and leave it missing otherwise. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (20)$$

be the set of populated cells; these become the calibration targets (helpers).

Coverage requirement. To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (21)$$

A day failing (21) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

5 The calibration as an optimization problem

Let $\mathcal{N}$ index the targets with market data $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$, all priced at the single spot S_{ref} . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^5 [\ell_j, h_j] \subset \mathbb{R}^5, \quad (\ell, h) \text{ in the } (\theta, \kappa, \eta, \rho, v_0) \text{ order of Table 2.} \quad (22)$$

5.1 Residuals

Two residual systems are used: a price-space system that the local optimizer minimises, and an IV-space system that selects among restarts and gates acceptance.

Price-space (in-engine objective). For each target the market value is the Black price at the quoted vol, $C_n^{\text{mkt}} = C_B(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$, and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Phi) = \frac{C_H(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (23)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Phi) = \left(\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Phi)^2 \right)^{1/2}. \quad (24)$$

This is the only error type QuantLib’s helpers expose, and is what Levenberg–Marquardt drives to a local minimum.

IV-space (selection and gate). Invert each Heston price back to a Black vol via (11) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Phi) &= \mathcal{I}(C_H(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Phi) &= \sigma_n^{\text{mod}}(\Phi) - \hat{\sigma}_n, \end{aligned} \quad (25)$$

with objective

$$J_{\text{iv}}(\Phi) = \left(\frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Phi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Phi) \text{ inverts finitely}\}. \quad (26)$$

J_{iv} is measured in volatility points; it is the natural surface-fit metric and, unlike J_{px} , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

5.2 Multi-start local optimization

A single start into a non-convex problem is fragile, so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right), \quad (27)$$

and define the seed set $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$ of $(v_0, \kappa, \theta, \eta, \rho)$ tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (28)$$

For each seed $s \in \mathcal{S}$ we solve the box-constrained nonlinear least-squares problem

$$\hat{\Phi}_s = \text{LM-local}\left(\min_{\Phi \in \mathcal{B}} |\mathcal{N}| J_{\text{px}}(\Phi)^2\right) \text{ started at } \Phi^{(0)} = s, \quad (29)$$

using Levenberg–Marquardt under the box constraint $\mathcal{B}$ (`QuantLib NonhomogeneousBoundaryConstraint`). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Phi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Phi}_s). \quad (30)$$

Ranking by J_{iv} (not J_{px}) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

6 Acceptance gate

Φ^* is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction δ , be

$$\text{PEGGED}(\Phi) = \bigvee_{j=1}^5 \left[(\Phi_j - \ell_j) < \delta (h_j - \ell_j) \quad \vee \quad (h_j - \Phi_j) < \delta (h_j - \ell_j) \right]. \quad (31)$$

The day is accepted iff

$$\boxed{J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Phi^*)}. \quad (32)$$

The first clause is the IV-space gate ($\varepsilon_{\text{iv}} \approx 2$ vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of J_{iv} . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

Why an IV-space gate. An earlier version gated on J_{px} directly. On a pooled day rich in deep-OTM wings, J_{px} is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with J_{iv} aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is ≈ 0.8 – 1.4 vol points, and the residual rejections are boundary-pegged κ or ρ (a separate identification lever), not fit-quality failures.

7 Algorithm

Algorithm 1 states the per-day routine; the driver applies it to every trading day and accumulates accepted rows into a single file (§8).

Algorithm 1 Per-day Heston calibration

Require: day d of OTM trades; rates $r(d), g(d)$; constants of Table 1

- 1: $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$ (pre-filter (13))
- 2: **if** $r(d)$ or $g(d)$ undefined on/before d **then**
- 3: **return** $\emptyset$ (no rate)
- 4: **end if**
- 5: $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$; $R \leftarrow \max_i S_i / \min_i S_i - 1$; $\text{HIGH_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6: $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$ with $m_i = K_i / S_i$ (re-strike (16))
- 7: $\mathcal{T} \leftarrow$ top- N_t maturities by volume; build $\mathcal{K}_\tau$ (n_k per wing); pivot to $\{\hat{\sigma}_{k,\tau}\}$, cells $\mathcal{N}$
- 8: **if** coverage (21) fails **then**
- 9: clear stale diagnostics; **return** $\emptyset$
- 10: **end if**
- 11: $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$; build seeds $\mathcal{S}$ (28)
- 12: **for** $s \in \mathcal{S}$ **do**
- 13: $\hat{\Phi}_s \leftarrow$ LM-local minimiser of J_{px} on $\mathcal{B}$ from s (29)
- 14: compute $J_{\text{iv}}(\hat{\Phi}_s)$ (26)
- 15: **end for**
- 16: $\Phi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Phi}_s)$ (select (30))
- 17: **if** $J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}}$ **and** $\neg \text{PEGGED}(\Phi^*)$ **then**
- 18: write repriced diagnostics; **return** $\text{row}(d, S_{\text{ref}}, r, g, \Phi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20: clear stale diagnostics; **return** $\emptyset$ (rejected)
- 21: **end if**

8 Output and diagnostics

An accepted day yields one parameter row keyed by date,

$$(d, S_{\text{ref}}, r, g, \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*, \mathcal{F}(\Phi^*), J_{\text{iv}}, J_{\text{px}}, \dots, R, \text{HIGH_MOVE}), \quad (33)$$

together with coverage counts ($|\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|$) and total volume. Because the calibration is once-daily, these rows accumulate into a single file, regenerated each run from the accepted days. For repricing diagnostics each retained contract is valued under Φ^* at its *original* (S_i, K_i) — not (S_{ref}, K^*) — since the structural parameters are spot-independent; this gives an honest comparison against the actual trade price P_i .

9 Constants

Symbol	Code name	Value	Role
$\tau_{\min}$	MIN_DTM	7 days	minimum maturity (drop ultra-short)
$R_{\max}$	MAX_MOVE_PCT	0.03	intraday-range flag threshold
Δ	STRIKE_GRID	5.0	SPX strike lattice for K^* snap
N_t	MAX_NT	12	maturities kept (by volume)
n_k	MAX_NK	8	strikes kept per wing
M_t	MIN_MATS	3	minimum maturities in surface
M_k	MIN_STRIKES	5	minimum strikes in surface
M_{cell}	MIN_CELLS	12	minimum populated cells $ \mathcal{N} $
ε_{iv}	IV_RMSE_ACCEPT	0.02	IV-space acceptance gate (vol pts)
δ	BOUND_TOL	10^{-3}	boundary-pegging tolerance fraction

Table 1: Surface, selection, and gate constants.

	θ	κ	η	ρ	v_0
lower ℓ	10^{-4}	10^{-2}	10^{-2}	-0.999	10^{-4}
upper h	1.0	20.0	2.0	0.5	1.0

Table 2: Box bounds $\mathcal{B}$, in the `model.params()` order $(\theta, \kappa, \eta, \rho, v_0)$. The ρ upper bound is kept slightly positive (equity leverage implies $\rho < 0$) and η is capped at 2.0 (SPX vol-of-vol ~ 0.3 – 1.2).